

St George Girls High School

Year 12

Mid-HSC Course Examination

2015



Mathematics

Extension 1

General Instructions

- Working time – 90 minutes
- Reading time – 5 minutes
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.

Total marks – 65

Section I: 5 marks

- Attempt Questions 1 – 5
- All questions are of equal value
- Use the multiple choice answer sheet provided

Section II : 60 marks

- Attempt Questions 6 – 10
- All questions are of equal value
- In Questions 6 – 10, show relevant mathematical reasoning and/or calculations

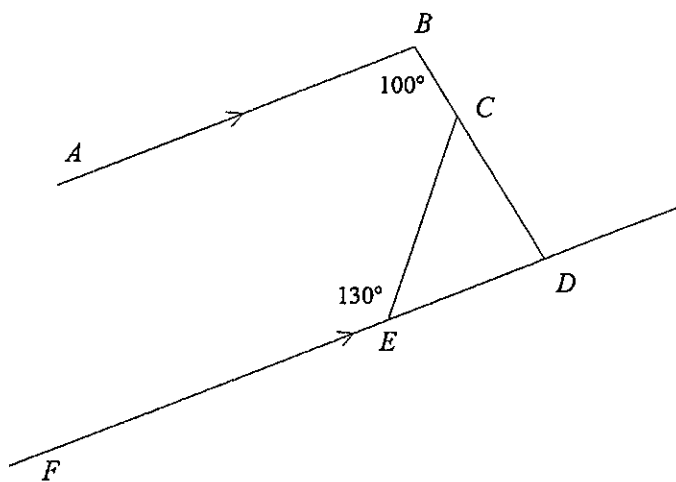
Section I:

5 Marks

Attempt Questions 1 – 5

Use the multiple choice answer sheet provided for Questions 1 – 5.

1. In the diagram below, AB is parallel to FD , $\angle ABC = 100^\circ$ and $\angle CEF = 130^\circ$



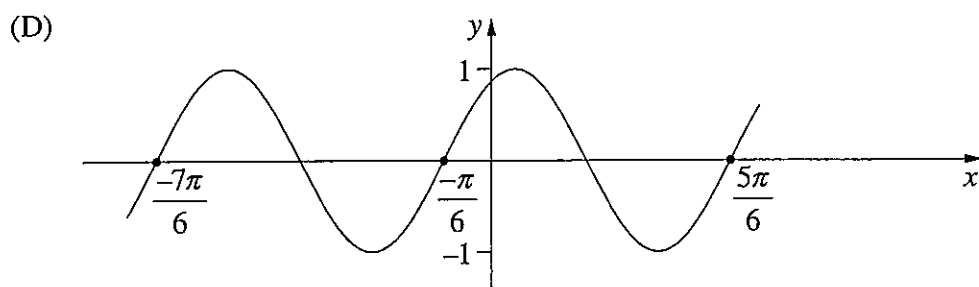
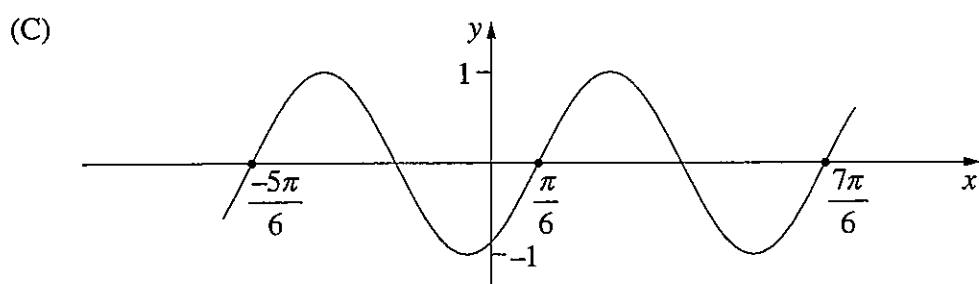
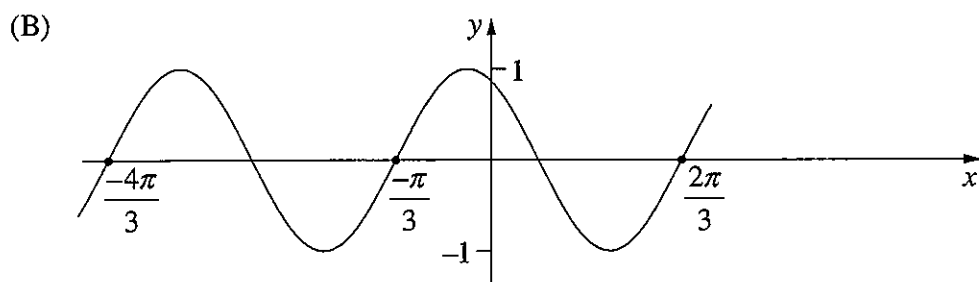
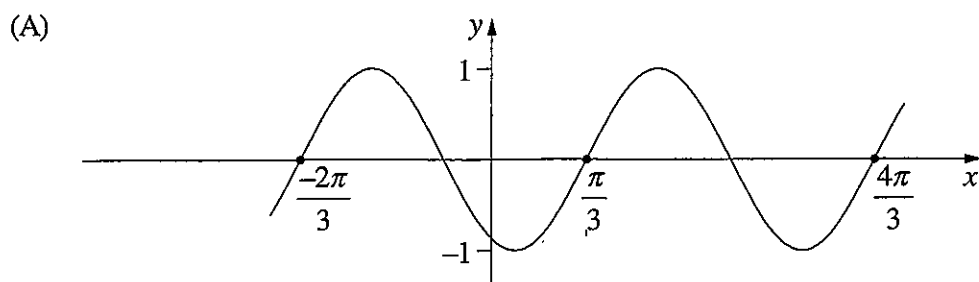
What is the value of $\angle BCE$?

- (A) 100°
(B) 110°
(C) 120°
(D) 130°
2. Which of the following expressions is correct?

- (A) $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1-x^2}}$
(B) $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$
(C) $\tan^{-1} x = \cos^{-1} \frac{x}{\sqrt{1-x^2}}$
(D) $\tan^{-1} x = \cos^{-1} \frac{x}{\sqrt{1+x^2}}$

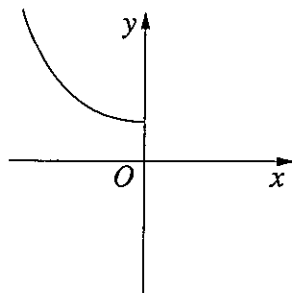
Section I (cont'd)

3. Which diagram shows the graph $y = \sin\left(2x + \frac{\pi}{3}\right)$?

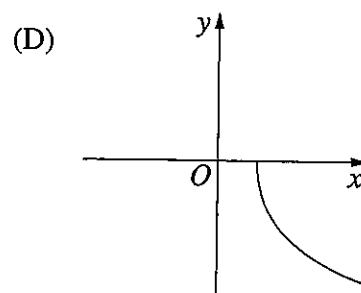
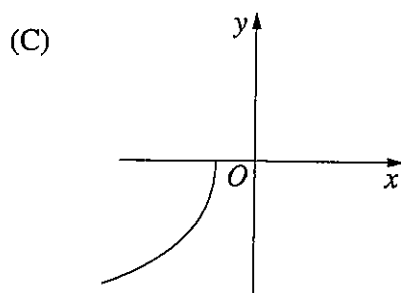
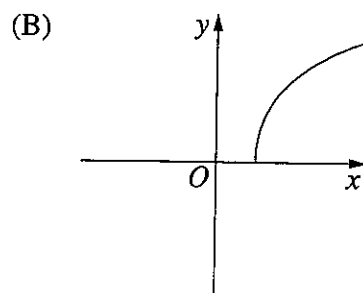
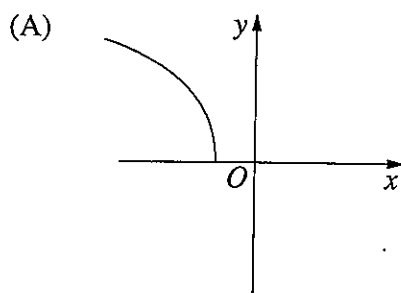


Section I (cont'd)

4. The diagram shows the graph $y = f(x)$.



Which diagram shows the graph $y = f^{-1}(x)$?



5. What is the primitive function of $3 \sin^2 \frac{2x}{5}$?

(A) $\frac{3x}{2} - \frac{6}{5} \sin \frac{4x}{5} + c$

(B) $\frac{3x}{2} - \frac{15}{8} \sin \frac{4x}{5} + c$

(C) $3x - \frac{15}{4} \sin \frac{4x}{5} + c$

(D) $3x - \frac{6}{5} \sin \frac{4x}{5} + c$

End of Section I

Section II

60 marks

Attempt Questions 6 – 10

Start each question in a new booklet.

In Questions 6, 7, 8, 9 and 10 your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (12 Marks) Use a SEPARATE writing booklet.

Marks

a) Let $f(x) = \sqrt{4x - 3}$.

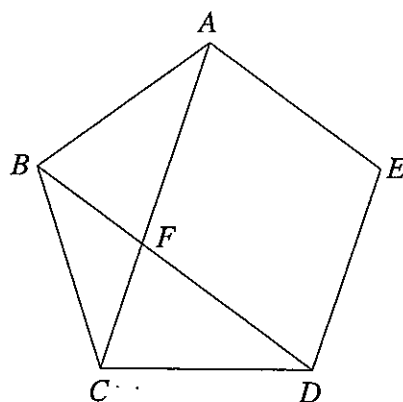
(i) Find the domain of $f(x)$. 1

(ii) Find an expression for the inverse function $f^{-1}(x)$. 2

(iii) Find the points where the graphs $y = f(x)$ and $y = x$ intersect. 2

(iv) On the same set of axes, sketch the graphs $y = f(x)$ and $y = f^{-1}(x)$ showing the information found in part (iii). 2

b)



NOT
TO
SCALE

In the diagram, $ABCDE$ is a regular pentagon. The diagonals AC and BD intersect at F .

Copy or trace this diagram into your writing booklet.

(i) Show that the size of $\angle ABC$ is 108° . 1

(ii) Find the size of $\angle BAC$. Give reasons for your answer. 2

(iii) By considering the sizes of angles, show that $\triangle ABF$ is isosceles. 2

Question 7 (12 Marks) Use a SEPARATE writing booklet. **Marks**

a) Show that $\sin \left(2 \tan^{-1} \frac{1}{2} \right) = \frac{4}{5}$. 2

b) Given that $f(x) = 3 \cos^{-1} \left(\frac{x-1}{2} \right)$.

(i) Write down the domain and range of $y = f(x)$. 2

(ii) Draw a neat sketch of $y = f(x)$, showing all important features. 2

c) The region bounded by $y = \cos 4x$ and the x -axis, between $x = 0$ and $x = \frac{\pi}{8}$, is rotated about the x -axis to form a solid.

Find the volume of the solid. 3

d) Prove that $\frac{\sin x - \cos 2x + 1}{\sin 2x + \cos x} = \tan x$. 3

Question 8 (12 Marks) Use a SEPARATE writing booklet.

Marks

- a) ABC is an isosceles triangle with a right-angle at B . The sides AB, BC are each of length 2 cm. An arc, centre A , radius 2 cm cuts the side AC at D .

(i) Draw a diagram to represent all this information. 1

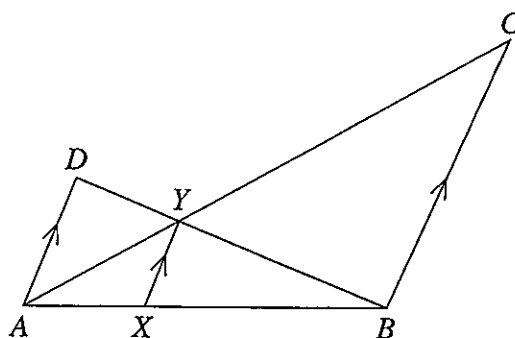
(ii) If BDC is the part of $\triangle ABC$ outside the circle, show that the area of BDC is $\left(2 - \frac{\pi}{2}\right) \text{ cm}^2$. 2

- b) Find $\lim_{x \rightarrow 0} \frac{\tan 5x}{\sin \frac{1}{3}x}$. 2

You must justify your answer.

- c) Find all solutions of $\sin(\pi + x) = \tan x$. 3

- d) The diagram shows triangles ABC and ABD with AD parallel to BC . The sides AC and BD intersect at Y . The point X lies on AB such that XY is parallel to AD and BC .



(i) Prove that $\triangle ABC$ is similar to $\triangle AXY$. 2

(ii) Hence, or otherwise, prove that $\frac{1}{XY} = \frac{1}{AD} + \frac{1}{BC}$. 2

Question 9 (12 Marks) Use a SEPARATE writing booklet. **Marks**

a) Differentiate $x^2 \sin^{-1} 5x$. 2

b) Use the t -formula to solve $2 \sin \theta + \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$. 3

c) (i) Show that $\frac{\sec^2 \theta}{\tan \theta} = \frac{1}{\sin \theta \cos \theta}$. 1

(ii) Find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\sin \theta \cos \theta} d\theta$. 2

d) (i) Show that $\cos \theta - \sqrt{3} \sin \theta = 2 \cos \left(\theta + \frac{\pi}{3} \right)$. 2

(ii) Hence, solve $\cos \theta - \sqrt{3} \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$. 2

Question 10 (12 Marks) Use a SEPARATE writing booklet.

Marks

- a) If $\sin \alpha = \frac{3}{4}$ and $\sin \beta = \frac{2}{3}$ where $0 < \alpha < \frac{\pi}{2}$, and $\frac{\pi}{2} < \beta < \pi$, find the exact value of $\cos(\alpha - \beta)$.

4

- b) If $f(x) = 2 \cos^{-1} \left(\frac{x}{\sqrt{2}} \right) - \sin^{-1}(1 - x^2)$

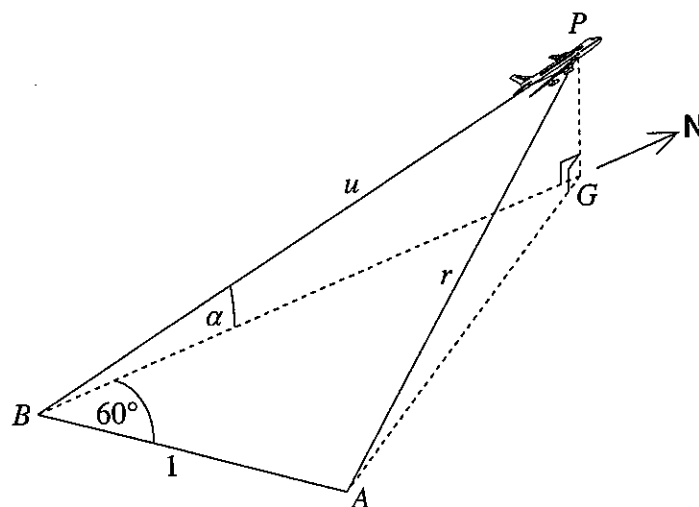
(i) Find $f'(x)$.

3

(ii) Hence, prove that $f(x) = \frac{\pi}{2}$.

2

- c) A plane P takes off from a point B . It flies due north at a constant angle α to the horizontal. An observer is located at A , 1 km from B , at a bearing 060° from B . Let u km be the distance from B to the plane and let r km be the distance from the observer to the plane. The point G is on the ground directly below the plane.



Show that $r = \sqrt{1 + u^2 - u \cos \alpha}$.

3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Student Name: _____

Class Teacher: _____

Section I

Year 12 Mid-HSC Course

Mathematics Extension 1

Multiple-choice Answer Sheet - Questions 1 – 5

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

 A ○ B ● C ○ D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐

- | | | | | | | | | |
|----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

2015 HSC MATHEMATICS EXTENSION 1 MID-COURSE EXAMINATION SOLUTIONS

1 $\angle CED = 50^\circ$
 $\angle CDE = 80^\circ$
 $\angle ECD = 50^\circ$
 $\therefore \angle BCE = 130^\circ$

$\therefore \underline{D}$

2 Let $\theta = \tan^{-1} x$
 $x = \tan \theta$

$\therefore \cos \theta = \frac{1}{\sqrt{1+x^2}}$

$\theta = \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$

$\therefore \tan^{-1}(x) = \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \therefore \underline{B}$

3 x -intercepts when $\sin \left(2x + \frac{\pi}{3} \right) = 0$

$2x + \frac{\pi}{3} = 0$

$2x = -\frac{\pi}{3}$

$x = -\frac{\pi}{6}$

$\therefore \underline{D}$

4 $y = f^{-1}(x)$ is a reflection of $y = f(x)$ in the line $y = x \therefore \underline{D}$

5 $\int 3 \sin \left(\frac{2x}{5} \right)^2 dx = 3 \int \left(\frac{1}{2} - \frac{1}{2} \cos \frac{4x}{5} \right) dx$
 $= 3 \left[\frac{1}{2} x - \frac{5}{8} \sin \frac{4x}{5} \right] + C$
 $= \frac{3}{2} x - \frac{15}{8} \sin \frac{4x}{5} + C \therefore \underline{B}$

Question 6

a) i $4x - 3 \geq 0$

$4x \geq 3$

$x \geq \frac{3}{4}$

$\therefore$ Domain: all real x , $x \geq \frac{3}{4}$

ii let $y = \sqrt{4x-3}$
 inverse $x = \frac{y^2+3}{4}$

$4y = x^2+3$

$y = \frac{x^2+3}{4}$

$\therefore f^{-1}(x) = \frac{x^2+3}{4}$

$x^2 = 4y - 3$

iii Points of intersection where $\sqrt{4x-3} = x$
 $4x-3 = x^2$

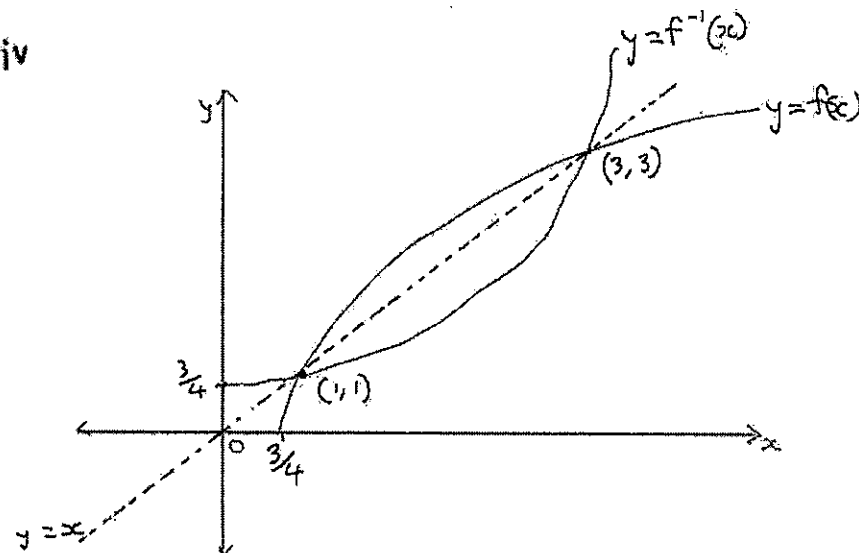
$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

$\therefore x = 3, x = 1$

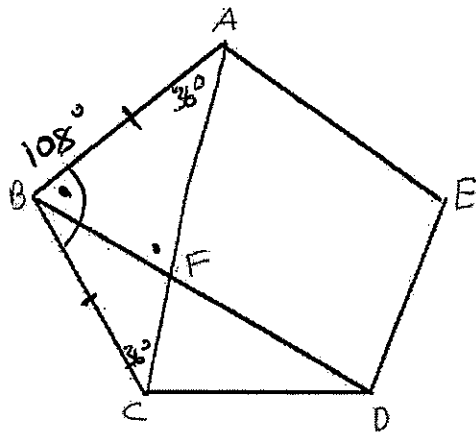
$\therefore$ points are (1,1) and (3,3)

iv



Question 6 continued

b



$$\begin{aligned} \text{i } \angle ABC &= \frac{(5-2) \times 180}{5} \\ &= \frac{540}{5} \\ &= 108^\circ \end{aligned}$$

$$\begin{aligned} \text{ii } AB &= BC \text{ (equal sides of regular pentagon)} \\ \therefore \angle BAC &= \angle BCA \text{ (equal angles opposite equal sides)} \\ 2\angle BAC + 108^\circ &= 180^\circ \text{ (angle sum of } \triangle BAC) \\ 2\angle BAC &= 72^\circ \\ \angle BAC &= 36^\circ \end{aligned}$$

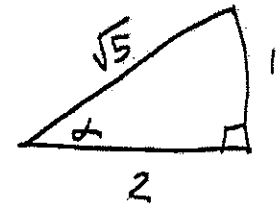
$$\begin{aligned} \text{iii } \text{Similarly, } \triangle BCD &\text{ is isosceles, and } \angle CBD = 36^\circ \\ \angle ABF + \angle CBD &= 108^\circ \\ \angle ABF &= 108 - 36 \\ &= 72^\circ \end{aligned}$$

$$\begin{aligned} \text{Also } \angle AFB + 72^\circ + 36^\circ &= 180^\circ \text{ (angle sum of } \triangle AFB) \\ \therefore \angle AFB &= 72^\circ \\ \text{So } \angle AFB &= \angle ABF = 72^\circ \\ \therefore \triangle ABF &\text{ is isosceles} \end{aligned}$$

Question 7

a

$$\begin{aligned} \text{Let } \alpha &= \tan^{-1} \frac{1}{2} \\ \therefore \tan \alpha &= \frac{1}{2} \end{aligned}$$



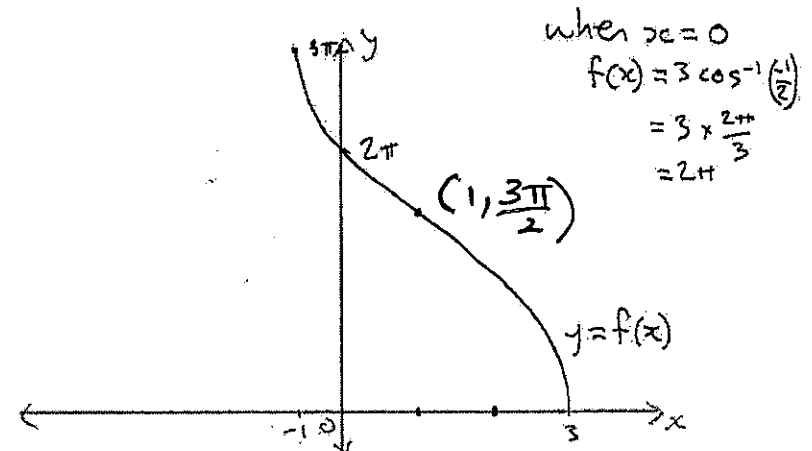
$$\begin{aligned} \sin \left(2 \tan^{-1} \frac{1}{2} \right) &= \sin 2\alpha \\ &= 2 \sin \alpha \cos \alpha \\ &= 2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} \\ &= \frac{4}{5} \end{aligned}$$

b i

$$\begin{aligned} \text{Domain of } \cos^{-1} x & \quad -1 \leq x \leq 1 \\ \text{so for } \cos^{-1} \left(\frac{x-1}{2} \right) & \quad -1 \leq \frac{x-1}{2} \leq 1 \\ & \quad -2 \leq x-1 \leq 2 \\ & \quad -1 \leq x \leq 3 \end{aligned}$$

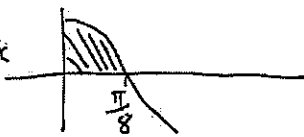
ii

$$\text{Range: } 0 \leq y \leq 3\pi$$



Question 7 continued

c) $V = \pi \int y^2 dx$

$$y^2 = \cos^2 4x = \frac{1}{2} + \frac{1}{2} \cos 8x$$


$$= \pi \int \left(\frac{1}{2} + \frac{1}{2} \cos 8x \right) dx$$

$$= \frac{\pi}{2} \left[x + \frac{\sin 8x}{8} \right]_0^{\pi/8}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{8} + 0 \right) - (0 + 0) \right]$$

$$= \frac{\pi^2}{16} \text{ units}^3$$

d) LHS = $\frac{\sin x - \cos 2x + 1}{\sin 2x + \cos x}$

$$= \frac{\sin x - (\cos^2 x - \sin^2 x) + 1}{2 \sin x \cos x + \cos x}$$

$$= \frac{\sin x - (1 - \sin^2 x) + \sin^2 x + 1}{\cos x (2 \sin x + 1)}$$

$$= \frac{2 \sin^2 x + \sin x}{\cos x (2 \sin x + 1)}$$

$$= \frac{\sin x (2 \sin x + 1)}{\cos x (2 \sin x + 1)}$$

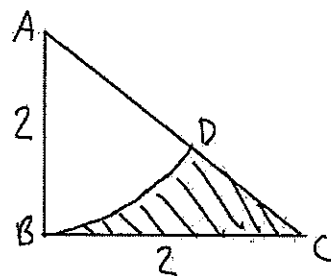
$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

$$= \text{RHS}$$

Question 8

a) i



$\angle BAC = 45^\circ$, as $AB = BC$

ii Area_{BDC} = Area_{ABC} - Area_{ABD}

$$= \frac{1}{2} \times 2 \times 2 - \pi \times 2^2 \times \frac{45}{360}$$

$$= 2 - \frac{4\pi}{8}$$

$$= 2 - \frac{\pi}{2} \text{ cm}^2$$

b) $\lim_{x \rightarrow 0} \frac{\tan 5x}{\sin \frac{1}{3}x} = \lim_{x \rightarrow 0} \frac{\tan 5x}{x} \times \frac{x}{\sin \frac{1}{3}x}$

$$= \lim_{x \rightarrow 0} 5 \times \frac{\tan 5x}{5x} \times 3 \times \frac{\frac{1}{3}x}{\sin \frac{1}{3}x}$$

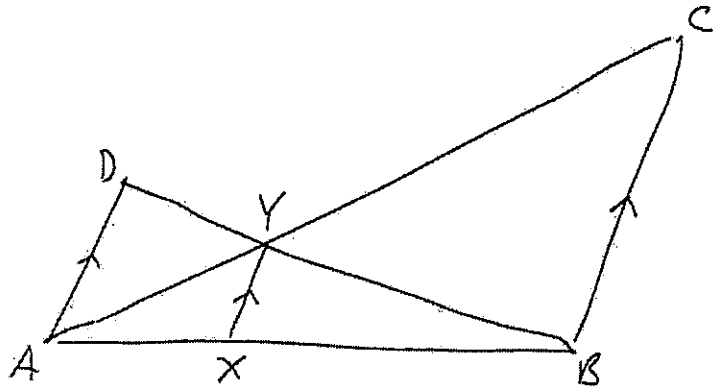
$$= 5 \times 1 \times 3 \times 1$$

$$= 15$$

c) $\sin(\pi + x) = -\sin x$ or $\cos x = -1$
 $-\sin x = \tan x$ $x = \pi, 3\pi, 5\pi, \dots$
 $-\sin x \cos x = \sin x$
 $\sin x + \sin x \cos x = 0$ $\therefore x = n\pi, n \in \mathbb{Z}$
 $\sin x (1 + \cos x) = 0$
 $\therefore \sin x = 0$ $x = 0, \pi, 2\pi, \dots$

Question 8 continued

d)



- i In triangles AXY and ABC :
 $\angle AXY = \angle ABC$ (corresponding angles, $XY \parallel BC$)
 $\angle YAX$ is common
 $\therefore \triangle ABC \sim \triangle AXY$ (equiangular)
- ii Similarly, $\triangle BAD \sim \triangle BXY$

$$\frac{XY}{AX} = \frac{XB}{AB} \text{ (corresponding sides of similar triangles)}$$

$$\frac{YX}{CB} = \frac{AX}{AB} \text{ (similarly)}$$

$$\text{so } \frac{XY}{AD} + \frac{XY}{BC} = \frac{AX}{AB} + \frac{XB}{AB}$$

$$XY \left(\frac{1}{AD} + \frac{1}{BC} \right) = \frac{AX+XB}{AB} = \frac{AB}{AB}$$

$$\therefore \frac{1}{AD} + \frac{1}{BC} = \frac{1}{XY}$$

Question 9

a) let $y = x^2 \sin^{-1} 5x$ let $u = x^2$ $v = \sin^{-1} 5x$
 $u' = 2x$ $v' = \frac{1}{\sqrt{1-25x^2}}$

$$y' = uv' + u'v = 2x \sin^{-1} 5x + \frac{x^2}{\sqrt{1-25x^2}}$$

b) $2 \sin \theta + \cos \theta = 1$ $0 \leq \theta < 2\pi$
 let $t = \tan \frac{1}{2} \theta$ $0 \leq \frac{1}{2} \theta < \pi$

$$\therefore 2 \left(\frac{2t}{1+t^2} \right) + \left(\frac{1-t^2}{1+t^2} \right) = 1$$

$$4t + 1 - t^2 = 1 + t^2$$

$$2t^2 - 4t = 0$$

$$2t(t-2) = 0$$

$$\therefore t = 0$$

$$t-2=0$$

$$\tan \frac{1}{2} \theta = 0$$

$$t = 2$$

$$\frac{1}{2} \theta = 0, \pi$$

$$\tan \frac{1}{2} \theta = 2$$

$$\theta = 0, 2\pi$$

$$\frac{1}{2} \theta = \tan^{-1} 2$$

$$\theta = 2 \tan^{-1} 2$$

$$= 2.214297$$

$$\approx 2.2 \text{ (1 d.p.)}$$

$$\therefore \theta = 0, 2.2, 2\pi$$

Check $\theta = \pi$

$$2 \sin \pi + \cos \pi = 1$$

$$-1 \neq 1$$

$$\therefore \theta = \pi$$

is not a solution.

c) i LHS = $\frac{\sec^2 \theta}{\tan \theta}$

$$= \frac{1}{\cos^2 \theta} \times \frac{\cos \theta}{\sin \theta}$$

$$= \frac{1}{\sin \theta \cos \theta}$$

$$= \text{RHS}$$

Question 9 continued

c) ii

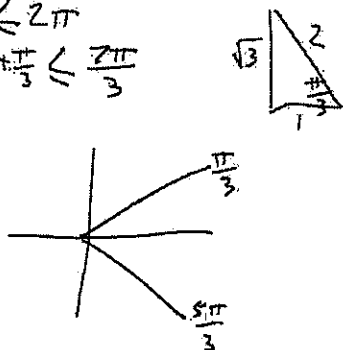
$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\sin \theta \cos \theta} d\theta &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 \theta}{\tan \theta} d\theta \\ &= \left[\ln(\tan \theta) \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\ &= \ln \sqrt{3} - \ln \frac{1}{\sqrt{3}} \\ &= \ln \sqrt{3} - \ln \sqrt{3}^{-1} \\ &= \ln \sqrt{3} + \ln \sqrt{3} \\ &= \ln 3 \end{aligned}$$

d) i R.H.S. = $2 \cos\left(\theta + \frac{\pi}{3}\right)$
 $= 2 \cos \theta \cos \frac{\pi}{3} - 2 \sin \theta \sin \frac{\pi}{3}$
 $= 2 \times \cos \theta \times \frac{1}{2} - 2 \times \sin \theta \times \frac{\sqrt{3}}{2}$
 $= \cos \theta - \sqrt{3} \sin \theta$
 $= \text{R.H.S.}$

ii $\cos \theta - \sqrt{3} \sin \theta = 1$ $0 \leq \theta \leq 2\pi$
 $2 \cos\left(\theta + \frac{\pi}{3}\right) = 1$ $\frac{\pi}{3} \leq \theta + \frac{\pi}{3} \leq \frac{7\pi}{3}$
 $\cos\left(\theta + \frac{\pi}{3}\right) = \frac{1}{2}$

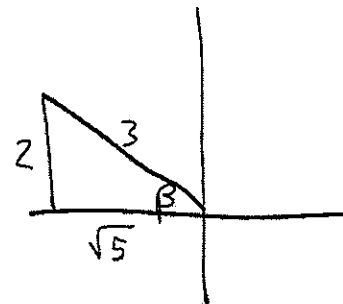
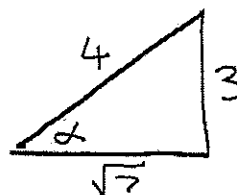
$$\therefore \theta + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\therefore \theta = 0, \frac{4\pi}{3}, 2\pi$$



Question 10

a)



$$\begin{aligned} \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\ &= \frac{\sqrt{2}}{4} \times \frac{\sqrt{5}}{3} + \frac{3}{4} \times \frac{2}{3} \\ &= \frac{\sqrt{10}}{12} + \frac{6}{12} \\ &= \frac{6 + \sqrt{10}}{12} \end{aligned}$$

b) i $f(x) = 2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1-x^2)$

$$\begin{aligned} f'(x) &= 2 \times \frac{-1}{\sqrt{2-x^2}} + \frac{-2}{\sqrt{2-x^2}} \\ &= 0 \end{aligned}$$

ii Since $f'(x) = 0$, $f(x)$ is a constant function, so $f(x) = k$, for some constant k .
 $f(0) = 2 \cos^{-1}\left(\frac{0}{\sqrt{2}}\right) - \sin^{-1}(1-0^2)$
 $= 2 \times \frac{\pi}{2} - \frac{\pi}{2}$
 $= \frac{\pi}{2}$

$$\therefore f(x) = \frac{\pi}{2}$$

Let $u = 1-x^2$

$$\frac{du}{dx} = -2x$$

$$y = \sin^{-1}(u)$$

$$y' = \frac{1}{\sqrt{1-u^2}}$$

$$= \frac{1}{\sqrt{1-(1-x^2)^2}}$$

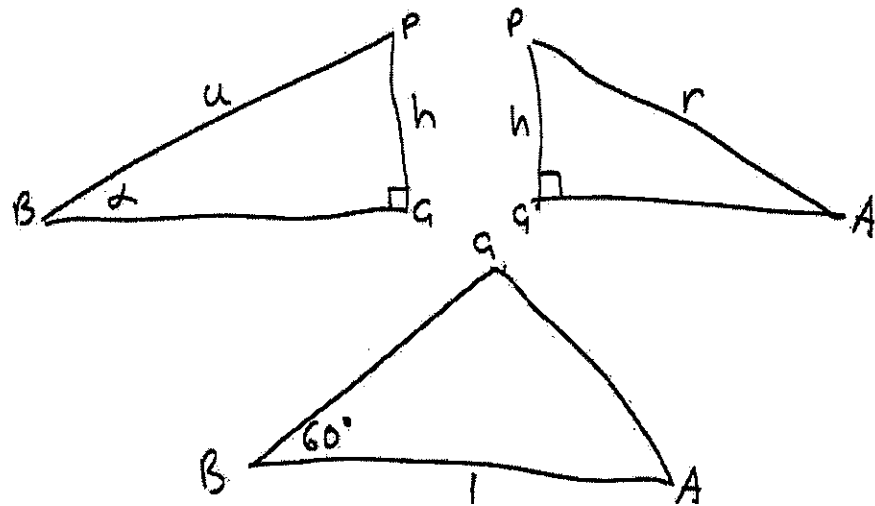
$$= \frac{1}{\sqrt{(1-(1-x^2))(1+(1-x^2))}}$$

$$= \frac{1}{\sqrt{x^2(2-x^2)}}$$

$$\therefore \frac{dy}{dx} = \frac{-2}{\sqrt{2-x^2}}$$

Question 10 continued

②



From $\triangle PGB$ $\cos \alpha = \frac{BG}{u}$
 $BG = u \cos \alpha$

also $BG^2 = u^2 - h^2$ (Pythagoras' Theorem)

From $\triangle PGA$ $r^2 = h^2 + AG^2$ (similarly)
 $AG^2 = r^2 - h^2$

From $\triangle ABC$
 $AG^2 = BG^2 + 1^2 - 2 \times BG \times 1 \times \cos 60$

$$r^2 - h^2 = u^2 - h^2 + 1 - 2 \times BG \times \frac{1}{2}$$

$$r^2 = 1 + u^2 - u \cos \alpha$$

$$r = \sqrt{1 + u^2 - u \cos \alpha}$$

Question 6. (12 marks)

(a) (i) $4x - 3 \geq 0$
 $4x \geq 3$
 $x \geq \frac{3}{4}$

D: $x \geq \frac{3}{4}$

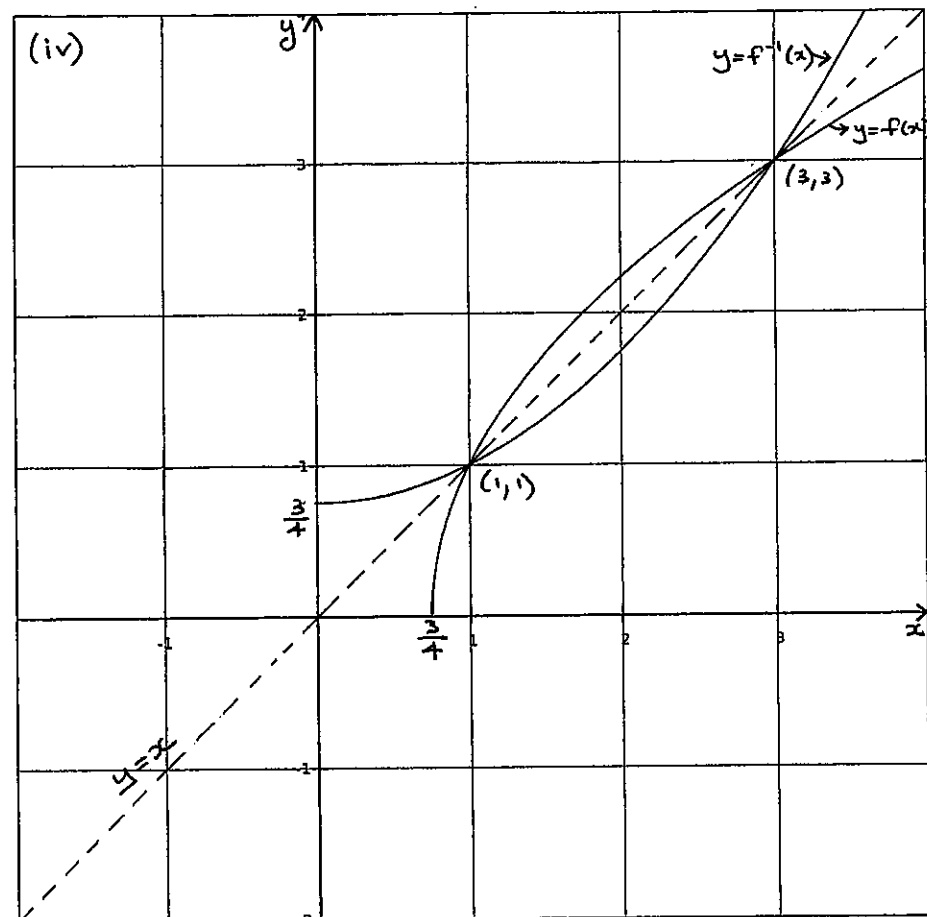
(ii) let $y = \sqrt{4x-3}$
 $x = \frac{y^2+3}{4}$
 $x^2 = \frac{y^2+3}{4}$
 $4y = x^2 + 3$
 $y = \frac{x^2+3}{4}$

$\therefore f^{-1}(x) = \frac{x^2+3}{4} = \frac{x^2}{4} + \frac{3}{4}$

(iii) $y = \sqrt{4x-3}$
 $y = x$

ie $x = \sqrt{4x-3}$
 $x^2 = 4x-3$
 $x^2 - 4x + 3 = 0$
 $(x-3)(x-1) = 0$
 $x = 3 \quad x = 1$
 $y = 3 \quad y = 1$

$\therefore (1, 1) \text{ \& } (3, 3)$



(b) (i) $\hat{A}BC = \frac{(5-2) \times 180^\circ}{5}$ (interior angles of a regular pentagon are equal).

$$= \frac{540^\circ}{5}$$

$\therefore \hat{A}BC = 108^\circ$

(ii) $AB = BC$ (equal sides of a regular polygon)

$\hat{B}AC = \hat{B}CA$ (angles opposite equal sides are equal)

$2\hat{B}AC + 108^\circ = 180^\circ$ (angle sum of a triangle is 180°)

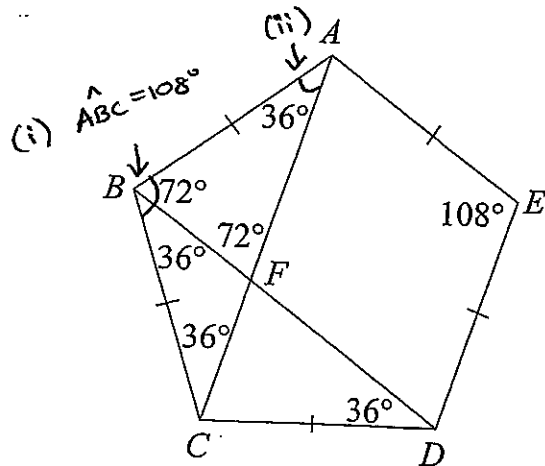
$2\hat{B}AC = 72^\circ$

$\therefore \hat{B}AC = 36^\circ$

OR

$$\therefore \hat{B}AC = \frac{180^\circ - 108^\circ}{2}$$

$$= 36^\circ$$



(iii) Similarly, $\triangle BDC$ is isosceles and $\hat{C}BD = \hat{C}DB = 36^\circ$

$\hat{A}BC = \hat{A}BF + \hat{C}BD = 108^\circ$

$\hat{A}BF = 108^\circ - 36^\circ$
 $= 72^\circ$

$\hat{A}FB + 72^\circ + 36^\circ = 180^\circ$ (angle sum of triangle ABF is 180°)

$\hat{A}FB = 180^\circ - 108^\circ$

$\hat{A}FB = 72^\circ$

$\therefore \hat{A}BF = \hat{A}FB$ (both equal to 72°)

$\therefore AB = AF$ (sides opposite equal angles are equal).

$\therefore \triangle ABF$ is isosceles

OR
(ii) In ΔBAC ,

$AB = BC$ (equal sides of a regular pentagon)

$\therefore \Delta BAC$ is isosceles

$\hat{BAC} = \hat{BCA}$ (equal angles of isosceles triangle)

$2\hat{BAC} + 108^\circ = 180^\circ$ (angle sum of a triangle is 180°)

$$2\hat{BAC} = 72^\circ$$

$$\therefore \hat{BAC} = 36^\circ \quad \#$$

OR

(iii) In Δ s ABC & BCD

$BA = BC$ (equal sides of a regular pentagon)

$\hat{ABC} = \hat{BCD}$ (interior angles of regular pentagon are equal)
 $= 108^\circ$

$BC = CD$ (equal sides of a regular pentagon)

$\therefore \Delta ABC \equiv \Delta BCD$ (SAS)

$\therefore \hat{BAC} = \hat{CBD}$ (corresponding angles in congruent triangles are equal)
 $= 36^\circ$

$\hat{BAC} = \hat{BCA}$ (equal angles in isosceles ΔBAC)
 $= 36^\circ$

$\hat{BFA} = \hat{CBD} + \hat{BCA}$ (exterior angle equals the sum of the two interior opposite angles)
 $= 36^\circ + 36^\circ$
 $= 72^\circ$

$$\begin{aligned}\hat{ABF} &= \hat{ABC} - \hat{CBD} \\ &= 108^\circ - 36^\circ \\ &= 72^\circ\end{aligned}$$

$\therefore \hat{BFA} = \hat{ABF}$ (both equal to 72°)

$\therefore AB = AF$ (sides opposite equal angles are equal)
 $\therefore \Delta ABF$ is isosceles